LAB ASSIGNMENT 6.2

2303A52488

TASK 1: STUDENT CLASS FOR PASS/FAIL

PROMPT USED: GENERATE A PYTHON STUDENT CLASS WITH NAME, ROLLNO, MARKS ATTRIBUTES. INCLUDE INIT AND IS_PASS METHOD (PASS IF MARKS >=40). GENERATED CODE:

```
CLASS STUDENT:
    DEF __INIT__(SELF,
NAME, ROLLNO,
MARKS):
SELF.NAME = NAME
SELF.ROLLNO =
ROLLNO
SELF.MARKS = MARKS
DEF IS_PASS(SELF):
IF SELF.MARKS >= 40:


RETURN
TRUE
```

```
ELSE:
RETURN
FALSE

# TEST
S1 = STUDENT("JOHN DOE", "CS001", 45)
PRINT(F"{S1.NAME} ({S1.ROLLNO}):
{'PASS' IF S1.IS_PASS() ELSE 'FAIL'}")

S2 = STUDENT("JANE SMITH", "CS002", 35)
PRINT(F"{S2.NAME} ({S2.ROLLNO}):
{'PASS' IF S2.IS_PASS() ELSE 'FAIL'}")
```

OUTPUT:

```
JOHN DOE (CS001): PASS
JANE SMITH (CS002): FAIL
```

ANALYSIS: CONSTRUCTOR INITIALIZES ATTRIBUTES CORRECTLY. IS_PASS METHOD SIMPLE AND

ACCURATE. LOGIC CLEAR, NO ERRORS. GOOD CLASS DESIGN FOR VALIDATION.

TASK 2: STAR PATTERN WITH FOR LOOP

PROMPT USED: WRITE FUNCTION

PRINT_TRIANGLE_FOR(N) TO PRINT

RIGHT-ANGLED TRIANGLE OF STARS

USING FOR LOOP. GENERATED CODE:

```
DEF PRINT_TRIANGLE_FOR(N):     FOR I IN
RANGE(N):         FOR J IN
```

```
RANGE(I+1):                PRINT("*", END="")
PRINT()


PRINT_TRIANGLE_FOR(5)
```

OUTPUT:

```
*

**

***

****

*****
```

PROMPT USED: SAME

PATTERN USING WHILE

LOOP. GENERATED CODE:

```
DEF PRINT_TRIANGLE_WHILE(N):
    I = 0     WHILE I < N:         J = 0
WHILE J <= I:             PRINT("*", END="")
J += 1         PRINT()         I += 1


PRINT_TRIANGLE_WHILE(5)
```

OUTPUT:

```
*
**
```

```
***

****

*****
```

ANALYSIS: BOTH LOOPS PRODUCE CORRECT PATTERN. FOR LOOP MORE CONCISE, WHILE LOOP

EXPLICIT. CONDITIONS PROPER, NO INFINITE LOOPS.

TASK 3: NUMBER CLASSIFICATION

PROMPT USED: FUNCTION

CHECK_NUMBER(NUM) RETURNS

'POSITIVE', 'NEGATIVE', OR 'ZERO' USING

IFELIF-ELSE. TEST WITH 10, -5, 0.

GENERATED CODE:

```
DEF CHECK_NUMBER(NUM):     IF
NUM > 0:
       RETURN "POSITIVE"     ELIF NUM < 0:
RETURN "NEGATIVE"     ELSE:         RETURN
"ZERO"


PRINT(CHECK_NUMBER(10))
PRINT(CHECK_NUMBER(-5))
PRINT(CHECK_NUMBER(0))
PRINT(CHECK_NUMBER(3.14))
```

OUTPUT:

```
POSITIVE NEGATIVE ZERO POSITIVE
```

ANALYSIS: HANDLES ALL CASES PERFECTLY. ELIF STRUCTURE EFFICIENT. WORKS FOR INTEGERS AND

FLOATS.

TASK 4: NESTED CONDITIONALS FOR DISCOUNT

PROMPT USED: FUNCTION

CHECK_DISCOUNT(AGE, IS_MEMBER) FOR

DISCOUNT: AGE>=60 SENIOR 10%,

MEMBER EXTRA 5%. USE NESTED IF.

GENERATED CODE:

```
DEF CHECK_DISCOUNT(AGE,
IS_MEMBER):
DISCOUNT = 0
IF AGE >= 60:
DISCOUNT += 10
        PRINT("SENIOR
DISCOUNT APPLIED")
IF IS_MEMBER:
        DISCOUNT += 5
        PRINT("MEMBER
DISCOUNT APPLIED")
RETURN DISCOUNT
```

```
PRINT(F"DISCOUNT:
{CHECK_DISCOUNT(65, TRUE)}%")
PRINT(F"DISCOUNT:
{CHECK_DISCOUNT(45, TRUE)}%")
PRINT(F"DISCOUNT:
{CHECK_DISCOUNT(70, FALSE)}%")
```

OUTPUT:

```
SENIOR DISCOUNT APPLIED
MEMBER DISCOUNT
APPLIED  DISCOUNT: 15%
MEMBER DISCOUNT
APPLIED  DISCOUNT: 5%
```

```
SENIOR DISCOUNT APPLIED
DISCOUNT: 10%
```

ANALYSIS: NESTED LOGIC CLEAR - SENIOR FIRST, THEN MEMBER. ACCUMULATES DISCOUNTS CORRECTLY. DECISION FLOW EASY TO FOLLOW.

TASK 5: CIRCLE CLASS MATH OPERATIONS

PROMPT USED: CIRCLE CLASS WITH RADIUS, METHODS AREA() AND CIRCUMFERENCE(). USE PI=3.14159.

GENERATED CODE:

```
IMPORT MATH
```

```
CLASS CIRCLE:     DEF
__INIT__(SELF, RADIUS):
SELF.RADIUS = RADIUS
DEF AREA(SELF):
RETURN MATH.PI *
SELF.RADIUS ** 2
DEF
CIRCUMFERENCE(SELF):
RETURN 2 * MATH.PI *
SELF.RADIUS


C = CIRCLE(7)
PRINT(F"AREA: {C.AREA():.2F}")
```

```
PRINT(F"CIRCUMFERENCE:
{C.CIRCUMFERENCE():.2F}")
```

OUTPUT:

```
AREA: 153.94
CIRCUMFERENCE: 43.98
```

ANALYSIS: FORMULAS CORRECT ($A=\pi R^2$, $C=2\pi R$). USES MATH.PI FOR PRECISION. METHODS WELL-STRUCTURED, OUTPUT FORMATTED NICELY.